

Replay-Aware CVRP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: cvrp_failure_replay.
- Best CVRPLIB holdout gap: cvrp_failure_replay.
- Best synthetic holdout gap: cvrp_random_replay.

Run Metadata

- run_name: run_20260513_191024_c
- started_at_local: 2026-05-13 19:10:24
- finished_at_local: 2026-05-13 19:13:14
- duration_hhmm: 00:03
- duration_seconds: 170.358
- seed_offset: 2000
- replicate_label: c
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
cvrp_no_replay	none	score_only	False	0.245759	0.012431	0.142058	0.908405	0.58	0.036692
cvrp_random_replay	random	score_only	False	0.251066	0.005376	0.14187	0.805416	0.58	0.039517
cvrp_failure_replay	failure	score_only	False	0.24297	0.005376	0.137373	0.889447	0.58	0.041784
cvrp_failure_replay_compression	failure	novelty_gate	True	0.253161	0.005376	0.143034	0.0	0.58	0.0

Condition Notes

cvrp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.245759, synthetic 0.012431, combined 0.142058.

- Accepted-epoch count 2, mean accepted code novelty 0.908405, and final complexity 0.58.
- Adaptation efficiency 0.036692 and archive sizes {'worst_cases': 5, 'failure_cases': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.245759 across 5 instances; family means: A=0.103839, B=0.269722, E=0.284069, P=0.301444.
- Panel synthetic_holdout mean gap 0.012431 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.028221.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 581, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.268559}, {"best_known_cost": 937, "cost": 1064, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.135539}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.251066, synthetic 0.005376, combined 0.14187.
- Accepted-epoch count 2, mean accepted code novelty 0.805416, and final complexity 0.58.
- Adaptation efficiency 0.039517 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.251066 across 5 instances; family means: A=0.103839, B=0.256232, E=0.333973, P=0.305054.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 577, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.259825}, {"best_known_cost": 937, "cost": 1057, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.128068}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.24297, synthetic 0.005376, combined 0.137373.
- Accepted-epoch count 2, mean accepted code novelty 0.889447, and final complexity 0.58.
- Adaptation efficiency 0.041784 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.24297 across 5 instances; family means: A=0.115183, B=0.256232, E=0.28215, P=0.305054.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.318777}, {"best_known_cost": 937, "cost": 1057, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.128068}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.

- Final transfer gaps: CVRPLIB 0.253161, synthetic 0.005376, combined 0.143034.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.253161 across 5 instances; family means: A=0.133508, B=0.256232, E=0.314779, P=0.305054.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.318777}, {"best_known_cost": 937, "cost": 1057, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.128068}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

Judge Appendix

Summary of CVRP Replay-Aware Benchmark Results

Condition	CVR PLIB Gap	Synthetic Holdout Gap	Transfer Gap	Code Novelty (Mean)	Replay Mode	Notes on Transfer & Novelty
cvrp_no_replay	0.2458	0.0124	0.1421	0.9084	None	Baseline with no replay; moderate gaps and highest code novelty. Transfer gap better than random replay but not best.
cvrp_random_replay	0.2511	**0.0054**	0.1419	0.8054	Random Replay	Lowest synthetic gap indicating best adaptation in synthetic holdout; transfer gap similar to no_replay; code novelty dropped but transfer not.
cvrp_failure_replay	**0.2430**	**0.0054**	**0.1374**	0.8894	Failure Replay	Best performance on CVRPLIB heldout and transfer gap metrics; synthetic holdout gap matches best (random replay). Code novelty slightly reduced compared to no_replay but higher than random replay.
cvrp_failure_replay_compression	0.2532	0.0054	0.1430	0.0	Failure Replay + Compression	Worst CVRPLIB and transfer gap among failure replay variants; identical synthetic gap. Zero code novelty likely due to compression pressure limiting code changes.

Interpretation

- ****Transfer Evidence****:
- *Failure replay* condition exhibits the lowest CVRPLIB heldout gap (0.2430) and transfer gap (0.1374), indicating it best supports transfer to unseen real-world data.
- *Random replay* excels at synthetic holdout (0.0054 gap) but shows slightly worse heldout gap (0.2511) than failure replay.
- No replay is intermediate in transfer gap (0.1421), lower than random replay despite higher code novelty.

- ****Code Novelty vs Transfer****:
- Code novelty is highest without replay (0.9084) but transfer metrics do not improve accordingly; higher novelty without replay does not translate into better transfer.
- Introducing replay (both failure and random) reduces code novelty (~0.8-0.9) yet improves or maintains transfer performance.
- Compression-aware failure replay drastically reduces code novelty to zero and degrades transfer, showing that novelty alone is not sufficient or beneficial under compression pressure.
- ****Replay Modes Impact****:
- *Failure replay* is the best replay method for transfer to heldout CVRPLIB instances, outperforming no replay and random replay in key deterministic metrics.
- *Random replay* provides best synthetic generalization but slightly worse real-world transfer, suggesting synthetic gap is not always aligned with CVRPLIB holds out generalization.
- Compression-aware replay harms transfer despite replaying failures, likely due to reduced adaptation efficiency (0 accepted epochs) and zero code novelty.
- ****Complexity and Adaptation****:
- All conditions maintain the same mean complexity (0.58), so complexity differences do not explain transfer variance.
- Adaptation efficiency is highest for failure replay (0.0418), next random replay (0.0395), lower no replay (0.0367), and zero for compression replay. This aligns with transfer gap improvements.

Key Takeaways

- Failure replay condition is the strongest performer on main transfer metrics (CVRPLIB gap and transfer gap), despite moderate decrease in code novelty compared to no replay.
- Random replay yields the best synthetic holdout gap but no significant improvement or even slightly worse CVRPLIB transfer gap than failure replay.
- Code novelty alone does not indicate algorithmic improvement or transfer performance; replay-based conditions show better transfer despite lower novelty.
- Compression-aware replay prevents adaptation and transfer improvement, highlighting a tradeoff when replay is combined with compression pressure.

****Conclusion:****

Failure replay is recommended for improved transfer in replay-aware CVRP training, as it achieves the lowest real-world CVRPLIB gap and transfer gap, showing reliable transfer gains that are not solely explained by code novelty increases.